

HKDSE Mathematics Compulsory Part — Paper 2 (Mock)

2026 · Paper 2 (MC) · 45 marks · 75 min

Section A

- Simplify $\frac{a^6b^{-1}}{a^2b^3}$.
A. $\frac{a^4}{b^2}$
B. $\frac{a^4}{b^4}$
C. $\frac{a^8}{b^4}$
D. a^4b^4
- Express 0.00000428 in scientific notation.
A. 4.28×10^{-5}
B. 42.8×10^{-7}
C. 4.28×10^6
D. 4.28×10^{-6}
- Solve $4(x - 1) = x + 8$.
A. $x = 3$
B. $x = 12$
C. $x = 6$
D. $x = 4$
- Simplify $\frac{x^2 - 25}{x^2 - 2x - 15}$.
A. $\frac{x-5}{x+3}$
B. $\frac{x+5}{x-3}$
C. $\frac{x+5}{x+3}$
D. $\frac{x-5}{x-3}$
- Factorize $64a^3 - 27$.
A. $(4a - 3)(16a^2 - 12a + 9)$
B. $(4a + 3)(16a^2 - 12a + 9)$
C. $(4a - 3)(16a^2 + 12a + 9)$
D. $(4a - 3)(4a^2 + 12a + 9)$
- If $y = \frac{3x+2}{x-4}$, express x in terms of y .
A. $x = \frac{4y-2}{y-3}$
B. $x = \frac{4y+2}{y+3}$
C. $x = \frac{4y+2}{y-3}$
D. $x = \frac{2-4y}{y-3}$
- Solve $2^{\{x\}} = 50$, correct to 3 significant figures.
A. 5.64
B. 5.65
C. 1.70
D. 25.0
- Let α and β be the roots of $4x^2 - 7x + 2 = 0$. Find $\alpha^2 + \beta^2$.
A. $\frac{49}{16}$
B. $\frac{41}{16}$
C. $\frac{25}{16}$
D. $\frac{33}{16}$
- If the equation $x^2 - kx + (2k + 4) = 0$ has two equal real roots, find k .
A. $k = -8$ or $k = 2$
B. $k = 8$ only
C. $k = 4$ or $k = -1$
D. $k = 8$ or $k = -2$

10. Solve $x^2 - x - 12 \leq 0$.

A. $x \leq -3$ or $x \geq 4$

B. $-4 \leq x \leq 3$

C. $-3 \leq x \leq 4$

D. $3 \leq x \leq 4$

11. The 2nd and 6th terms of an arithmetic sequence are 8 and 24 respectively. Find the sum of the first 12 terms.

A. 312

B. 300

C. 324

D. 156

12. The geometric sequence is 3, 6, 12, Find the 8th term.

A. 384

B. 192

C. 768

D. 256

13. Find the remainder when $3x^3 - x^2 + 2x - 5$ is divided by $x - 2$.

A. 19

B. 1

C. -5

D. 23

14. It is given that y varies inversely as x^2 . When $x = 4$, $y = 3$. Find the value of y when $x = 2$.

A. 6

B. $\frac{3}{4}$

C. 12

D. 48

15. The price of an item is increased by 10% twice successively. Find the overall percentage increase.

A. 21%

B. 20%

C. 121%

D. 11%

16. The sequence is 3, 7, 11, 15, Find the 15th term.

A. 59

B. 63

C. 55

D. 60

17. \$20000 is deposited at 4% per annum compounded yearly for 2 years. Find the amount, correct to the nearest dollar.

A. \$21600

B. \$21632.00

C. \$21632

D. \$20800

18. Simplify $\sqrt{75} - \sqrt{12}$.

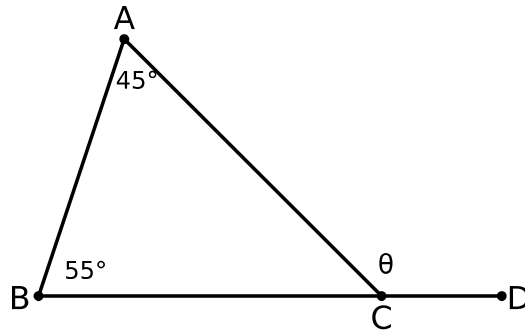
A. $\sqrt{3}$

B. $\sqrt{63}$

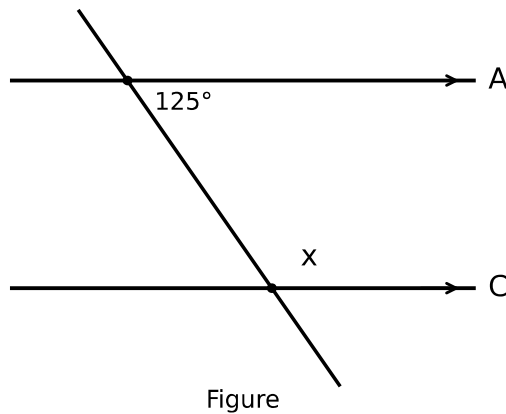
C. $3\sqrt{3}$

D. $5\sqrt{3}$

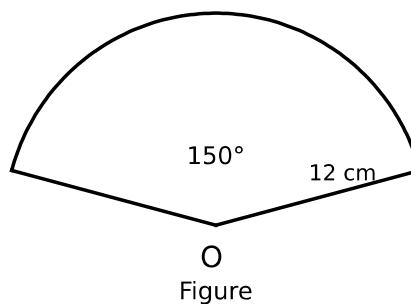
19. In the figure, in triangle ABC , $\angle A = 45^\circ$ and $\angle B = 55^\circ$. Find $\angle ACD$, where D lies on BC produced.



- A. 80°
B. 110°
C. 100°
D. 90°
20. In the figure, $\overline{AB} \parallel \overline{CD}$ and a transversal cuts them. The marked angle is 125° . Find x .

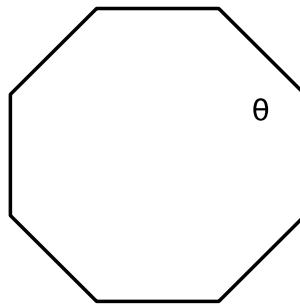


- A. 55°
B. 125°
C. 62.5°
D. 180°
21. In the figure, a sector has radius 12 cm and angle at the centre 150° . Find the length of the arc in terms of π .



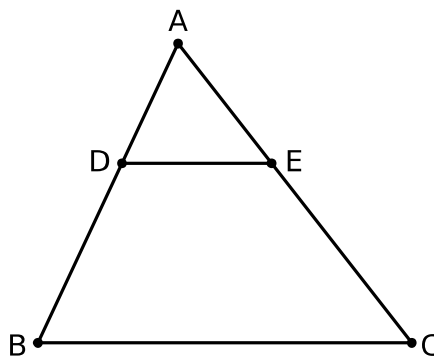
- A. 5π cm B. 10π cm
C. 60π cm D. 24π cm

22. Find the distance between the points (2, 6) and (6, 9) .
- A. 7
B. $\sqrt{7}$
C. 5
D. 25
23. Find the slope of a line perpendicular to the line $5x - 3y = 2$.
- A. $\frac{5}{3}$
B. $-\frac{3}{5}$
C. $\frac{3}{5}$
D. $-\frac{5}{3}$
24. Find the centre and radius of the circle $x^2 + y^2 - 8x + 6y - 24 = 0$.
- A. centre $(-4, 3)$, $r = 7$
B. centre $(4, -3)$, $r = 49$
C. centre $(8, -6)$, $r = 7$
D. centre $(4, -3)$, $r = 7$
25. The figure shows a regular decagon. Find each interior angle θ .



Figure

- A. 144°
B. 36°
C. 135°
D. 150°
26. In the figure, $DE \parallel BC$. It is given that $AD = 10$, $DB = 3$ and $AE = 10$. Find EC .



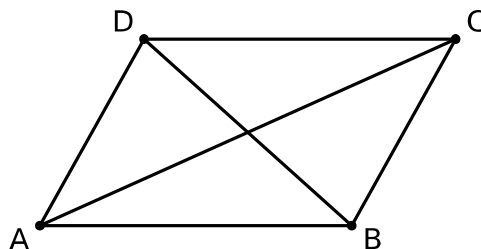
Figure

- A. $\frac{10}{3}$
B. 30
C. 3
D. 13
27. Find the volume of a sphere of radius 4 cm in terms of π .
- A. $\frac{64}{3}\pi \text{ cm}^3$
B. $\frac{256}{3}\pi \text{ cm}^3$
C. $64\pi \text{ cm}^3$
D. $16\pi \text{ cm}^3$

28. If the mean of 8, 10, x , 16 is 12, find x .
- A. 14
B. 12
C. 10
D. 13
29. Find the median of the data 3, 6, 6, 9, 11, 14.
- A. 6
B. 9
C. 7.5
D. 8
30. A fair die is thrown. Find the probability of getting a number greater than 4.
- A. $\frac{1}{2}$
B. $\frac{2}{3}$
C. $\frac{1}{6}$
D. $\frac{1}{3}$

Section B

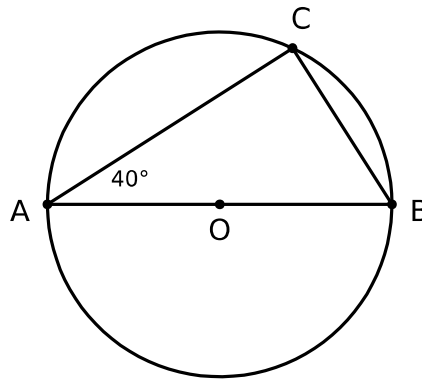
31. Let $a > b$, where a and b are real numbers. I. $a^2 > b^2$ II. $a - b > 0$ III. $-a < -b$ Which of the above is/are true?
- A. I and II only
B. I and III only
C. II and III only
D. I, II and III
32. Let x and y be positive numbers. I. $\log(x^2) = 2\log x$ II. $\log(x + y) = \log x + \log y$ III. $\log \frac{x}{y} = \log x - \log y$ Which of the above is/are true?
- A. I and II only
B. I and III only
C. II and III only
D. I, II and III
33. Consider the quadratic function $y = x^2 - 6x + 11$. I. The vertex of the graph is (3, 2) II. The graph opens upwards III. The graph cuts the x -axis at two points Which of the above is/are true?
- A. I and II only
B. I and III only
C. II and III only
D. I, II and III
34. Let ABCD be a parallelogram. I. The diagonals bisect each other II. All interior angles are equal III. Both pairs of opposite sides are parallel Which of the above is/are true?



Figure

- A. I and II only
B. I and III only
C. II and III only
D. I, II and III

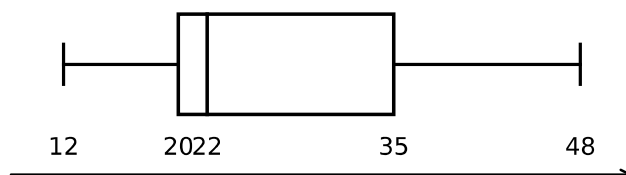
35. In the figure, AB is a diameter of the circle O and C is a point on the circle with $\angle BAC = 40^\circ$. Find $\angle ABC$.



Figure

- A. 50° B. 40°
C. 90° D. 80°
36. Consider the sequence 3, 6, 12, 24, I. It is a geometric sequence with common ratio 2
II. It is an arithmetic sequence III. The 6th term is 96 Which of the above is/are true?
A. **I and II only** B. **I and III only**
C. **II and III only** D. **I, II and III**
37. Each datum of a set of data is multiplied by 2 and then increased by 5 . If the standard deviation of the original data is 6 , find the standard deviation of the new data.
A. 17 B. 6
C. 29 D. 12
38. If $C^{\{n\}}_{\{2\}} = 66$, find n .
A. 11 B. 12
C. 66 D. 10
39. Let A and B be mutually exclusive events with $P(A) = 0.35$ and $P(B) = 0.45$. Find $P(A \cup B)$.
A. 0.8 B. 0.1575
C. 0.1 D. 0.63
40. 5 different books are arranged in a row such that 2 particular books must be next to each other. Find the number of different arrangements.
A. 240 B. 24
C. 120 D. 48
41. The nth term of a sequence is $T(n) = 2n^2 + n$. Find $T(6) - T(5)$.
A. 22 B. 78
C. 23 D. 55

42. In a test, the mean is 50 marks and the standard deviation is 12 marks. A student has a standard score of -1.5 . Find the student's marks.
- A. 68
B. 32
C. 48.5
D. 30
43. A box-and-whisker diagram shows: minimum 10, lower quartile 18, median 20, upper quartile 34, maximum 46. Which of the following about the distribution is correct?



- A. **negatively skewed** / 負偏 (左偏) B. **symmetric** / 對稱
- C. **IQR is 20** / 四分位數間距為 20 D. **positively skewed** / 正偏 (右偏)
44. A bag contains 4 red balls and 2 white balls. 2 balls are drawn at random at the same time. Find the probability of drawing at least 1 white ball.
- A. $\frac{3}{5}$ B. $\frac{2}{5}$
- C. $\frac{1}{3}$ D. $\frac{4}{5}$
45. Find the variance of the data 1, 3, 5, 7, 9.
- A. 4 B. $2\sqrt{2}$
- C. 40 D. 8

END OF PAPER